

## Day 13: Man Made Materials & Metallurgy

To Do:

- ✓ Materials like- Rubber (process like vulcanisation)
- ✓ Common chemicals like caustic potash, baking soda, etc and their uses
  - Petrol and other fuels
- ✓ Polymers and uses
  - Paints, Fertilisers, etc.
- Various separating techniques
- ✓ Metals and their ores
- ✓ Properties of metals and their applications
- ✓ Alloys and their uses

### Method of Separation of Mixtures

Different methods of separation are used depending on the type of mixture. Some commonly used methods for separation include:

- Sublimation
- Evaporation
- Handpicking
- Threshing
- Winnowing
- Sieving
- Distillation
- Filtration or Sedimentation
- Separating Funnel
- Magnetic Separation

### Important metals and their ores

| Metal          | Ores               | Chemical Formula                                                                       |
|----------------|--------------------|----------------------------------------------------------------------------------------|
| Sodium (Na)    | Chile saltpeter    | $\text{NaNO}_3$                                                                        |
|                | Trona              | $\text{Na}_2\text{CO}_3 \cdot 2\text{NaHCO}_3 \cdot 3\text{H}_2\text{O}$               |
|                | Borax              | $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$                           |
|                | Common salt        | $\text{NaCl}$                                                                          |
|                | Bauxite            | $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$                                      |
| Aluminium (Al) | Corundum           | $\text{Al}_2\text{O}_3$                                                                |
|                | Felspar            | $\text{KAlSi}_3\text{O}_8$                                                             |
|                | Cryolite           | $\text{Na}_3\text{AlF}_6$                                                              |
|                | Alunite            | $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 4\text{Al}(\text{OH})_3$ |
|                | Kaolin             | $3\text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2 \cdot 2\text{H}_2\text{O}$                 |
|                | Nitre (salt peter) | $\text{KNO}_3$                                                                         |
| Potassium (K)  | Carnalite          | $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$                             |
| Magnesium (Mg) | Magnesite          | $\text{MgCO}_3$                                                                        |
|                | Dolomite           | $\text{MgCO}_3 \cdot \text{CaCO}_3$                                                    |
|                | Epsom salt         | $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$                                              |
|                | Kieserite          | $\text{MgSO}_4 \cdot \text{H}_2\text{O}$                                               |
|                | Carnalite          | $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$                             |



IMPORTANT

|                |                |                                                                     |
|----------------|----------------|---------------------------------------------------------------------|
| Calcium (Ca)   | Dolomite       | $\text{CaCO}_3 \cdot \text{MgCO}_3$                                 |
|                | Calcite        | $\text{CaCO}_3$                                                     |
|                | Gypsum         | $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$                           |
|                | Fluorspar      | $\text{CaF}_2$                                                      |
|                | Asbestos       | $\text{CaSiO}_3 \cdot \text{MgSiO}_3$                               |
| Strontium (Sr) | Strontianite   | $\text{SrCO}_3$                                                     |
|                | Silestine      | $\text{SrSO}_4$                                                     |
| Copper (Cu)    | Cuprite        | $\text{Cu}_2\text{O}$                                               |
|                | Copper glance  | $\text{Cu}_2\text{S}$                                               |
|                | Copper pyrites | $\text{CuFeS}_2$                                                    |
| Silver (Ag)    | Ruby Silver    | $3\text{Ag}_2\text{S} \cdot \text{Sb}_2\text{S}_3$                  |
|                | Horn Silver    | $\text{AgCl}$                                                       |
| Gold (Au)      | Calaverite     | $\text{AuTe}_2$                                                     |
|                | Silvenites     | $[(\text{Ag}, \text{Au}) \text{Te}_2]$                              |
| Barium (Ba)    | Barytes        | $\text{BaSO}_4$                                                     |
| Zinc (Zn)      | Zinc blende    | $\text{ZnS}$                                                        |
|                | Zincite        | $\text{ZnO}$                                                        |
|                | Calamine       | $\text{ZnCO}_3$                                                     |
|                | Cinnabar       | $\text{HgS}$                                                        |
| Mercury (Hg)   | Cassiterite    | $\text{SnO}_2$                                                      |
| Tin (Sn)       | Galena         | $\text{PbS}$                                                        |
| Lead (Pb)      | Stibnite       | $\text{Sb}_2\text{S}_3$                                             |
| Antimony (Sb)  | Greenocite     | $\text{CdS}$                                                        |
| Cadmium (Cd)   | Bismuthite     | $\text{Bi}_2\text{S}_3$                                             |
| Bismuth (Bi)   | Haemetite      | $\text{Fe}_2\text{O}_3$                                             |
| Iron (Fe)      | Lemonite       | $2\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$                  |
|                | Magnetite      | $\text{Fe}_3\text{O}_4$                                             |
|                | Siderite       | $\text{FeCO}_3$                                                     |
|                | Iron Pyrite    | $\text{FeS}_2$                                                      |
|                | Copper Pyrites | $\text{CuFeS}_2$                                                    |
|                | Smelite        | $\text{CoAsS}_2$                                                    |
|                | Milarite       | $\text{NiS}$                                                        |
| Cobalt (Co)    | Pyrolusite     | $\text{MnO}_2$                                                      |
|                | Magnite        | $\text{Mn}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$                   |
| Nickel (Ni)    | Carnetite      | $\text{K}(\text{UO})_2 \cdot \text{VO}_4 \cdot 3\text{H}_2\text{O}$ |
|                | Pitch blende   | $\text{U}_3\text{O}_8$                                              |
| Magnese (Mn)   |                |                                                                     |
| Uranium (U)    |                |                                                                     |

**Metals** are the element that loses electrons easily and have positive ions or cations. It consists of metallic bonds.

They are shiny and ductile i.e. can be converted into thin wires.

They are malleable i.e. can be converted into thin sheets.

They are dense and very good conductor of heat and electricity.

They are lustrous and dense.

They are elastic and tough.

Metals can be converted into liquid so, it is fusible also.

**Ores** are those minerals from which metals are extracted commercially and economically. Like Bauxite ( $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ ) is the ore of Aluminium which contains 50% to 70% of Aluminium Oxide.

The meaning of the term 'alloy' is a substance formed from the combination of two or more metals. **Alloys** can also be formed from combinations of metals and other elements. The properties exhibited by alloys are often quite different from the properties of their individual components. It is not uncommon for alloys to have greater strength and hardness when compared to pure metals. An example of an alloy is red gold, which is produced by alloying copper and gold together. Another important alloy of gold is white gold, which is produced by alloying silver and gold together.

### Properties of Alloys

Each alloy has certain useful properties. An alloy's properties are distinct from those of the individual metals from which it is produced. Some properties of alloys are given below.

- 1 Alloys are harder than their constituent metals.
- 2 Alloys are more resistant to corrosion than pure metals.
- 3 Alloys are more durable than the metals they are made from.
- 4 The electrical conductivity of alloys is lower than that of pure metals.
- 5 Alloys have a lower melting point than the metals from which they are made.
- 6 Alloys have greater ductility than their constituent metals.

| SL. | ALLOYS          | COMPOSITION                  | USES                                                            |
|-----|-----------------|------------------------------|-----------------------------------------------------------------|
| 1   | Alnico          | Al- 12% Ni- 20% Co- 5%       | Permanent Magnets                                               |
| 2   | Bell Metal      | Cu- 80% Sn- 20%              | Bell, Utensils, Coins                                           |
| 3   | Bearing Metal   | Sn- 82% Sb- 14% Cu- 14%      | Bearings                                                        |
| 4   | Dutch Metal     | Cu- 80% Zn-20%               | Machine Parts                                                   |
| 5   | Bronze          | Cu- 88% Sn- 12%              | Coins, Idols, Utensils                                          |
| 6   | Brass           | Cu- 67% Zn- 33%              | Locks, gears, bearings, doorknobs, ammunition, pipes and valves |
| 7   | Constantan      | Cu- 60%, Ni- 40%             | Electrical Appliances                                           |
| 8   | Coinage Alloys  | Cu-75% Ni- 25%               | Coins                                                           |
| 9   | Duralumin       | Al- 94% Cu-4% Mg, Mn, Si- 2% | Pressure Cooker, Aeroplanes                                     |
| 10  | German silver   | Au- 60% Zn- 25% Ni-15%       | Utensils                                                        |
| 11  | Gun Metal       | Cu- 86% Sn- 10% Zn- 4%       | Machine parts                                                   |
| 12  | Invar           | Fe- 63% Ni- 36% C- 1%        | Watch pendulum                                                  |
| 13  | Magnesium       | Al- 85% Mg- 15%              | Aeroplane Frame                                                 |
| 14  | Monel Metal     | Cu- 30% Ni-70%               | Chemical resistant containers                                   |
| 15  | Nichrome        | Ni Fe Cr                     | Heater coils                                                    |
| 16  | Rolled Gold     | Cu- 90% Al-10%               | Ornaments                                                       |
| 17  | Gold            | Au-98% Cu- 2%                | Jewellery, Electronics                                          |
| 18  | Pewter/ Solder  | Sn- 75% Pb- 25%              | Soldering wire                                                  |
| 19  | Type metals     | Pb- 75% Sb- 20% Sn- 5%       | Typesetting, printing processes                                 |
| 20  | Stainless steel | Fe- 73% Cr- 18% C- 1% Ni- 8% | Automobile parts, utensils, Crockery                            |
| 21  | Steel           | Fe- 98% C- 2%                | Nails, Screws, Bridges, Reinforced Concrete                     |

| Chemical Name                | Common Name         | Chemical Formulae                                            | Their Uses                                                                                                                                                        |
|------------------------------|---------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminium                    | Aluminium           | Al                                                           | Heat resistant clothing, Cookware, Greenhouses & Window Frames, in the making of Aircraft, Lightweighted Vehicles and ladders etc.                                |
| Acetic Acid                  | White Vinegar       | CH <sub>3</sub> COOH + H <sub>2</sub> O                      | In Grocery Store and Photography Store.                                                                                                                           |
| Acetylsalicylic Acid         | Aspirin             | C <sub>9</sub> H <sub>8</sub> O <sub>4</sub>                 | Pharmacy: - Aspirin                                                                                                                                               |
| Ammonium Chloride            | Soldering Flux      | NH <sub>4</sub> Cl                                           | Hardware Store :- Plumbing, Soldering iron stuffs and used as Dark Chemical in Photography.                                                                       |
| Ascorbic acid                | Vitamin C           | C <sub>6</sub> H <sub>8</sub> O <sub>6</sub>                 | Vitamin C tablets.                                                                                                                                                |
| Argon                        | Argon Gas           | Ar                                                           | Welding and inert filler gas inside incandescent lighting equipments such as Bulbs, CFLs.                                                                         |
| Amylose                      | Cornstarch          | (C <sub>6</sub> H <sub>9</sub> O <sub>5</sub> ) <sub>n</sub> | In Grocery Store                                                                                                                                                  |
| Ammonium Phosphate           | Fertilizer          | (NH <sub>4</sub> ) <sub>3</sub> PO <sub>4</sub>              | Used as fertilizer in Gardens & Agricultural Lands                                                                                                                |
| Aluminium Sulphate           | Flocculating Powder | Al <sub>2</sub> (SO <sub>4</sub> ) <sub>3</sub>              | Household Uses: - Styptic Pencils, Gardening, Water Treatment, Paper Making and Dyeing Fabrics.                                                                   |
| Acetylene Gas                | Ethyne              | C <sub>2</sub> H <sub>2</sub>                                | Carburization, welding, Portable Lighting, Mass Spectrometer and Radiocarbon Dating                                                                               |
| Ammonium Hydroxide           | Ammonia             | NH <sub>3</sub> ; NH <sub>4</sub> OH                         | Fertilizers, Explosives, Hair Care, Plastics, Refrigerators, Auto-care:- tire inflaters, Metal Cleaners, polishes, Shaving Cream, Acne Treatments, Lotions , etc. |
| Aluminium potassium Sulphate | Alum                | KAl(SO <sub>4</sub> ) <sub>2</sub> .12H <sub>2</sub> O       | Water Purification, Flame Retardant, Skin Whitener, Some types of Toothpastes, Pickling Agent and in Styptic pencils.                                             |

|                       |                            |                                                               |                                                                                                                                                              |
|-----------------------|----------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ammonium Bromide      | Ammonium Bromide           | NH <sub>4</sub> Br                                            | Fireproofing of Wood, Corrosion inhibitors, Pharmaceutical preparations, Lithography and process engraving                                                   |
| Ammonium Carbonate    | Smelling Salts             | (NH <sub>4</sub> ) <sub>2</sub> CO <sub>3</sub>               | Leavening agents, Acidity regulator, Smelling Salts, Cough Syrup                                                                                             |
| Acetone               | Acetone                    | CH <sub>3</sub> COCH <sub>3</sub>                             | Solvent, Skin lightening Products, Bath & Fragrance products                                                                                                 |
| Ammonium Bi-Fluoride  | Ammonium hydrogen fluoride | NH <sub>4</sub> HF <sub>2</sub>                               | Glass Processing, Toilet Bowl Cleaner, Rust Stain remover, Car body Cleaner, etc.                                                                            |
| Ammonium Nitrate      | Ammonium Nitrate           | NH <sub>4</sub> NO <sub>3</sub>                               | Used in Fertilizers and Explosives.                                                                                                                          |
| Aluminium Hydroxide   | Alumina Hydrate            | Al(OH) <sub>3</sub>                                           | Excellent Fire Retardant, Preparation of tetanus injections, Skin Protectant, Flame resistant, Water purification & Sewage Treatment, etc.                   |
| Aluminium Oxide       | Alumina                    | Al <sub>2</sub> O <sub>3</sub>                                | Glass Making, Gas Purification, Plastics making, Blush, Lipstick and Nail polishing, Alcohol dehydration, Paints, Composite Fiber, Abrasion Protection, etc. |
| Ammonium Persulphate  | Etching Solution           | (NH <sub>4</sub> ) <sub>2</sub> S <sub>2</sub> O <sub>8</sub> | Used as radical initiators in the polymerization of certain Alkenes, Hair Bleach, etc.                                                                       |
| Ammonium Bi-carbonate | Baker's Ammonia            | (NH <sub>4</sub> )HCO <sub>3</sub>                            | Nitrogen Fertilizer, raising agents for cookies and crackers, Ceramic manufacturing, etc.                                                                    |